CS217 - Data Structures & Algorithm Analysis (DSAA)

Lecture #14

▶Depth First Search & Applications

Prof. Pietro S. Oliveto
Department of Computer Science and Engineering
Southern University of Science and Technology (SUSTech)

olivetop@sustech.edu.cn
https://faculty.sustech.edu.cn/olivetop

Reading: Chapter 20 and
I. Wegener. A simplified correctness proof for a well-known algorithm computing strongly connected components. Information Processing Letters 83(1), pages 17–19 – On Blackboard

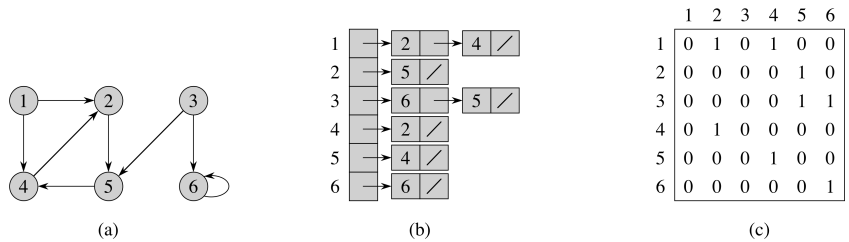
▶Aims for this lecture

- Introduce **depth-first search (DFS)** and depth-first trees.
- To show how DFS can **classify edges** for additional information about the graph.
- To show how to use DFS to
  - Check whether a graph contains cycles
  - Put tasks in the right order (topological sorting)
  - Compute strongly connected components in graphs
- To show the **correctness** of some remarkable algorithms.

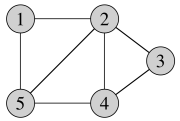
▶Representations of graphs

- Using terminology for graphs G = (V, E) from Appendix B
- **Adjacency-list representation**:
  - Array Adj of |V| lists, one for each vertex.
  - The list Adj[u] contains all vertices v adjacent to u in G, i.e. there is an edge $(u, v) \in E$.
  - The sum of all adjacency list lengths equals |E|.
- **Adjacency-matrix representation**:
  - Assume that vertices are numbered $1, 2, \ldots, n$.
  - Adjacency matrix is a $|V| \times |V|$ matrix with entries $a_{ij} = 1$ if $(i, j) \in E$ and $a_{ij} = 0$ otherwise.

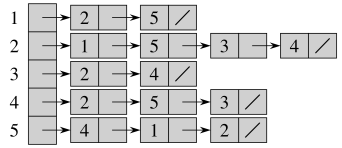
▶Example for a directed graph



▶Example for an undirected graph



(a)



(b)

	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0

(c)

- For every undirected edge $\{u, v\}$, v is in u's adjacency list and u is in v's adjacency list.
- Note the symmetry in the adjacency matrix along the main diagonal. It's sufficient to store the entries on and above the diagonal.

▶Depth-first search (DFS)

- Works for undirected and directed graphs.
- Ideas:
  - Go into depth by exploring edges out of the most recently discovered vertex and backtrack when stuck.
  - Continue until all vertices reachable from the start vertex are discovered.
  - If any undiscovered vertices remain, continue with one of them as new source.
- As for BFS, define predecessors $v.\pi$ that represent several **depth-first trees**.
- These trees form a **depth-first forest.**

Depth-First Tree (深度优先树)

定义：由所有的“前驱–后继”关系构成的边（称为树边，Tree Edges）所组成的结构。

Depth-First Forest (深度优先森林)

定义：由多棵“深度优先树”组成的集合。

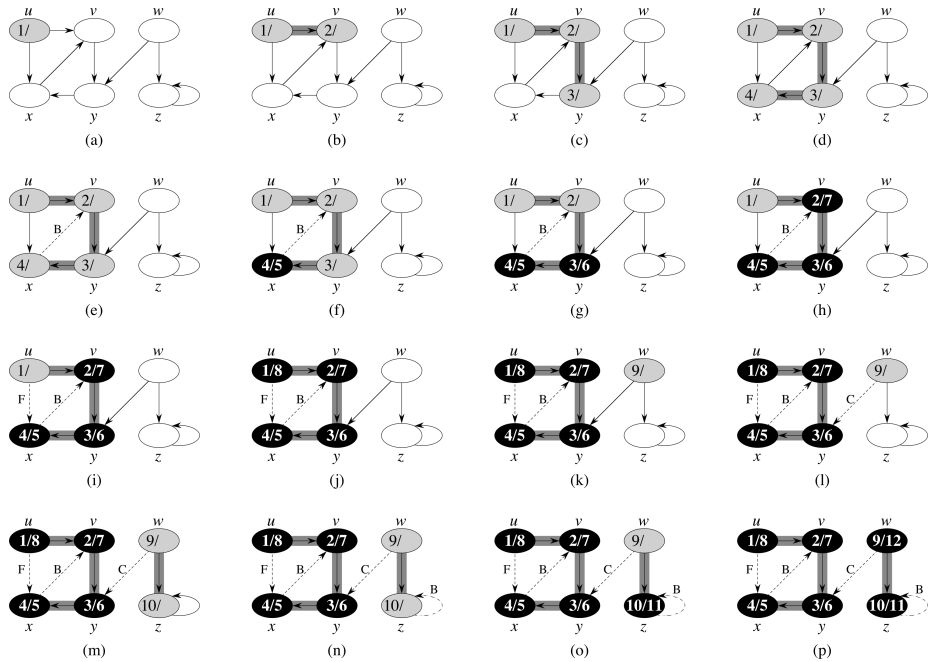
▶DFS: Colours and timestamps

- DFS uses colours white, gray, black as for BFS:
  - **White**: vertex has not been discovered yet
  - **Gray**: vertex has been discovered, but is not finished yet.
  - **Black**: vertex has been finished (finished scan of adjacency list).
- Also uses **timestamps**:
  - **v.d** is the time v is first **discovered** (and grayed)
  - **v.f** is the time v is **finished** (and blackened)
  - Global variable time is incremented with each event
  - Hence for all vertices $v.d < v.f$

▶DFS: Pseudocode

```
DFS(G)
for each vertex u ∈ V do
    u.colour = white
    u.π = NIL
time = 0
for each vertex u ∈ V do
    if u.colour == white then
        DFS-VISIT(G, u)
```

```
DFS-VISIT(G, u)
time = time+1
u.d = time
u.colour = gray
for each v ∈ Adj[u] do
    if v.colour == white then
        v.π = u
        DFS-VISIT(G, v)
u.colour = black
time = time+1
u.f = time
```



▶DFS: Pseudocode and runtime

```
DFS(G)
for each vertex u ∈ V do
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for each vertex u ∈ V do
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```

```
DFS-VISIT(G, u)
time = time+1
u.d = time
u.colour = gray
for each v ∈ Adj[u] do
    if v.colour == white then
        v.π = u
        DFS-VISIT(G, v)
u.colour = black
time = time+1
u.f = time
```

Runtime?

- Runtime is $\Theta(|V| + |E|)$:
  - DFS runs in time $\Theta(|V|)$ exclusive of the time for DFS-Visit.
  - DFS-Visit is only called once for each vertex v as *v* must be white and is grayed immediately. The loop executes |Adj[u]| times.
  - Since $\sum_{v \in V} |\mathrm{Adj}[v]| = \Theta(|E|)$, the total cost for loop is $\Theta(|E|)$.

▶Properties of DFS

Parenthesis structure: In any DFS of a (directed or undirected) graph, for any two vertices $u \neq v$, either

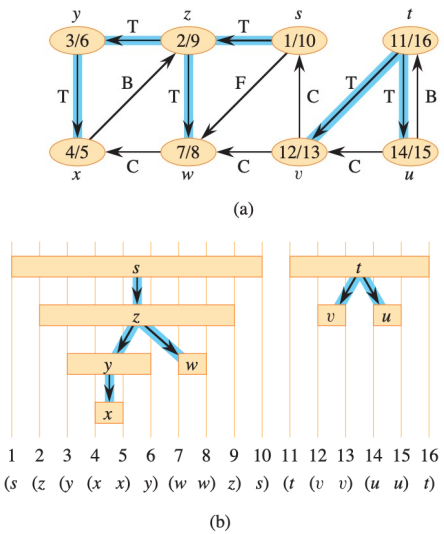
- DFS-Visit(v) is called during DFS-Visit(u), then v is a descendant of u and DFS-Visit(v) finishes earlier than u:

$$u.d < v.d < v.f < u.f$$

- the same happens with roles of v and u swapped, or
- the intervals $[u.d, u.f]$ and $[v.d, v.f]$ are entirely disjoint, and neither u nor v is a descendant of the other.

NB: Recursive calls mean that DFS implicitly uses a **stack** to store vertices while exploring the graph (cf. BFS using a queue).

▶Parenthesis structure: example



▶White-path theorem

Theorem 22.9: In a depth-first forest of a (directed or undirected) graph, vertex v is a descendant of a vertex u **if and only if** at the time $u.d$ that the search discovers u, there is a path from u to v consisting entirely of white vertices.

- This means: If and only if there is a white path from u to v, DFS will create a DFS tree with edges from u to v.
- "If and only if" indicates a statement like "$A \Leftrightarrow B$"
- We split this into two steps:
  1. Prove that $A \Rightarrow B$
  2. Prove that $A \Leftarrow B$
- It is often easier to focus on proving one implication.

descendant of u = 1. u = v; 2. v is after u;

▶White-path theorem (2)

Theorem 22.9: In a depth-first forest of a (directed or undirected) graph, vertex v is a descendant of a vertex u **if and only if** at the time $u.d$ that the search discovers u, there is a path from u to v consisting entirely of white vertices.

Proof of "⇒" (being descendant implies white path):

- If $u = v$ then u is still white when $u.d$ that is set, thus a white path from u to v exists (just one vertex $u = v$).
- If v is a proper descendant of u, then $u.d < v.d$ and therefore v is white at time $u.d$. This holds for all descendants of u, hence a white path from u to v exists at time $u.d$.

▶White-path theorem (3)

Theorem 22.9: In a depth-first forest of a (directed or undirected) graph, vertex v is a descendant of a vertex u **if and only if** at the time $u.d$ that the search discovers u, there is a path from u to v consisting entirely of white vertices.

Proof of "⇐" (white path implies descendancy) **by contradiction**:

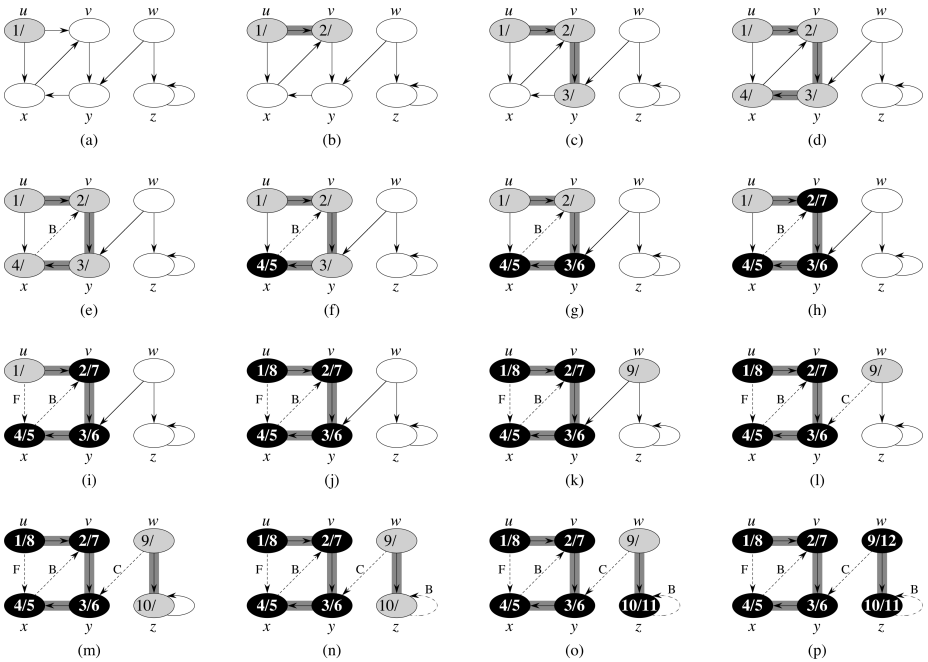
- Suppose there is a white path from u to v at time $u.d$.
- Assume v is the **first vertex on the path** which is not a descendant of u (otherwise we consider this first vertex instead).
- Let w be the predecessor of v on the path (could be $w = u$). Hence w must be a descendant of u (by above assumption). Thus $w.f \leq u.f$.
- v is discovered after u but before w is finished (as there is an edge from w to v), so we get: $u.d < v.d < w.f \leq u.f$.
- How large is $v.f$? Parenthesis structure tells us that $u.d < v.d < v.f < u.f$ is the only feasible case for $v.f$ and so v must be a descendant of u.

and we prove a contradiction
—> v is a descendant of u
—> all following not–a–descendant vertexes all lead t
contradiction
—> all descendants of u

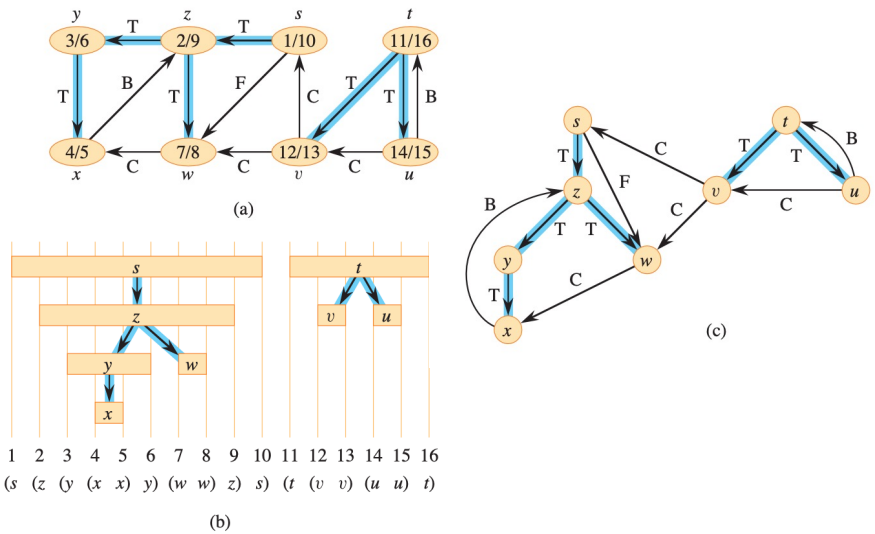
▶Classification of edges in directed graphs

DFS can be used to classify edges of the input graph.

1. **Tree edges** are edges in the depth-first forest. Edge (u, v) is a tree edge if, v was first discovered by exploring edge (u, v).
   *An edge (u, v) is a tree edge if at the time of exploration **v is white**.*
2. **Back edges** are edges (u, v) connecting a vertex u to an ancestor v in a depth-first tree (or self-loops in directed graphs).
   *An edge (u, v) is a back edge if at the time of exploration **v is gray**.*
3. **Forward edges** are nontree edges (u, v) connecting a vertex u to a descendant v in a depth-first tree (pointing forward in the tree).
   *(u, v) is a forward edge if **v is black and was discovered later**:*
   $u.d < v.d$.
4. **Cross edges** are all other edges: either leading to a subtree constructed earlier or leading to a different (earlier) depth-first tree.
   *(u, v) is a cross edge if **v is black and was discovered earlier**: $u.d > v.d$.*



▶Edge classification: example



▶Edge classification in undirected graphs

Theorem 22.10: In a depth-first search of an undirected graph, every edge is either a tree edge or a back edge.

→ There are no forward/cross edges in undirected graphs.

Proof:

- Let $\{u, v\}$ be an arbitrary edge, and assume without loss of generality that $u.d < v.d$.
- Since v is on u's adjacency list, search must discover and finish v before it finishes u.
- If the first time the edge is explored, it is in the direction from u to v, then v is undiscovered and it becomes a **tree edge**.
- If the edge is first explored from v to u, then it becomes a **back edge**, since u is still gray.

▶Precedence graphs

- Graphs have many applications. One of them is modelling precedences:
  - Vertices represent tasks
  - A edge (u, v) means that task u has to be executed before task v.
- Coming up: how to order tasks such that all precedence constraints are respected.
- But this is only feasible if the precedence graph does not contain any cycles!
- Such a graph is called **acyclic**.

有向图中，Cycle（环/回路） 的定义严格要求必须能够顺着箭头的方向（边的方向）走完一圈回到原点。

▶Application of DFS: testing for cycles

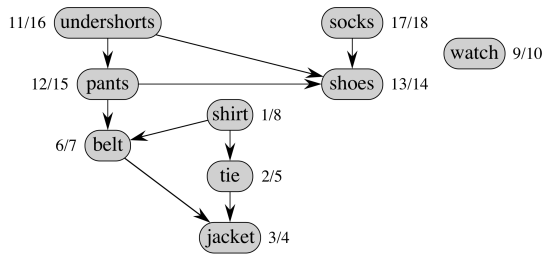
Theorem (adapted from Lemma 22.11): A graph G contains a cycle if and only if DFS finds at least one back edge.

Proof (for directed graphs):

- "⇐": Suppose DFS produces a back edge (u, v). Then v is an ancestor of u in the depth-first tree. Thus, G contains a path (of tree edges) from v to u, and the back edge completes a cycle.
- "⇒": Suppose that G contains a cycle C. We show that DFS yields a back edge. Let v be the first vertex to be discovered in C, and let (u, v) be the edge on C going into v. At time $v.d$, the vertices of C form a path of white vertices from v to u. By the white-path theorem, u becomes a descendant of v. Therefore, (u, v) is a back edge.

▶Topological sorting

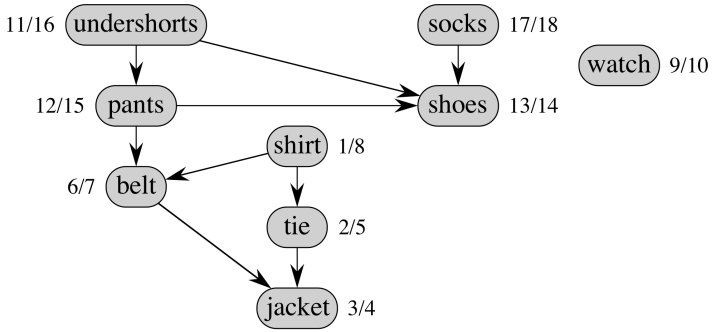
- Consider a directed acyclic graph ("dag").
- A topological sort of a dag is a linear ordering of all its vertices such that for each edge (u, v), u appears before v.
- If vertices are arranged on a horizontal line, all edges go from left to right.
- Example: Professor Bumstead getting dressed.



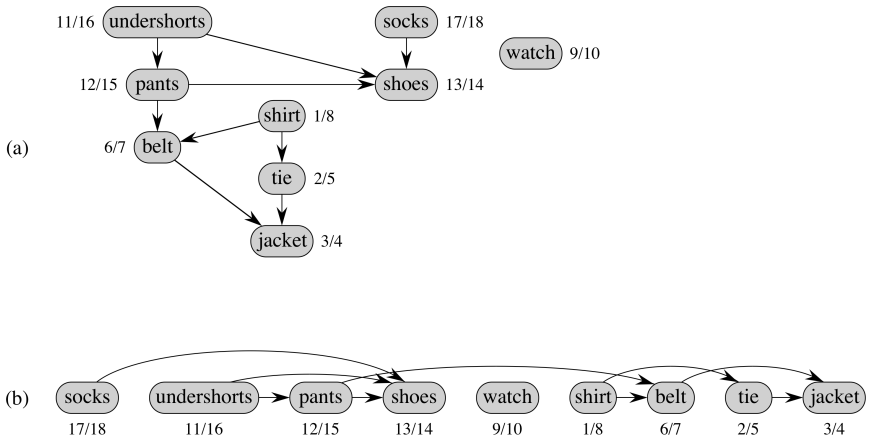
▶Computing a topological sort

- Here's how to use DFS to compute a topological sort:

```
TOPOLOGICAL-SORT(G)
call DFS(G) to compute finishing times v.f for each vertex v
as each vertex is finished, insert it onto the front of a linked list
return the linked list of vertices
```



▶Professor Bumstead getting dressed



▶Topological sort: Runtime

```
TOPOLOGICAL-SORT(G)
call DFS(G) to compute finishing times v.f for each vertex v
as each vertex is finished, insert it onto the front of a linked list
return the linked list of vertices
```

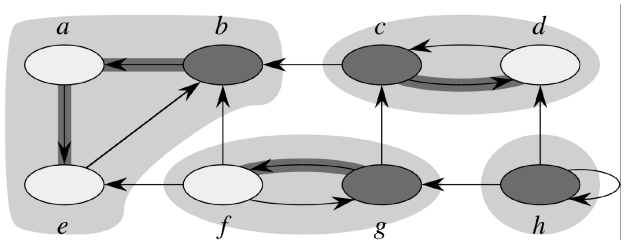
- Runtime:
  - $\Theta(|V| + |E|)$ time for DFS
  - $+O(1)$ for each vertex inserted in to the linked list $\rightarrow +O(|V|)$
  - Total time $\Theta(|V| + |E|)$
- Why on earth does this work?!

▶Topological sort: correctness proof

- Suffices to show that if G contains an edge (u, v), then $v.f < u.f$. Then v is inserted to the list earlier and will come to rest after u.
- Consider any edge (u, v) explored by DFS. When this edge is explored, v **cannot be gray**, since then v would be an ancestor of u and (u, v) would be a back edge, contradicting the fact that G is acyclic.
- Therefore, v must be either white or black.
  - If v is white, it becomes a descendant of u, and so $v.f < u.f$ by parenthesis structure.
  - If v is black, it has been finished and $v.f$ has been set. Because we are still exploring from u, a timestamp $u.f$ will be assigned later and once we do, it will be larger: $v.f < u.f$.

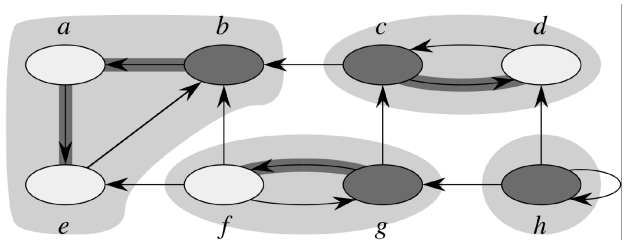
▶Strongly connected components

- A directed graph is called **strongly connected** if every two vertices are reachable from each other.
- The **strongly connected components (SCCs)** of a directed graph are the equivalence classes under the "mutually reachable" relation. In other words, they are maximal sets of vertices where all vertices in every set are mutually reachable.



▶Strongly connected components

- Applications:
  - Finding groups of friends in social network graphs. Instagram
  - Many algorithms working on directed graphs decompose the graph into its SCCs, run separately on all of them, and then combine solutions for all SCCs to one overall solution.



▶Computing SCCs with DFS

- Let $G^\top$ be the transpose of G, i. e. the graph where all edges have their direction reversed.
- Note that G and $G^\top$ have the same SCC as u and v are reachable in $G^\top$ if and only if they are reachable in G.
- $G^\top$ can be computed in time $O(|V| + |E|)$.

STRONGLY-CONNECTED-COMPONENTS(G)

1: call DFS(G) to compute finishing times $v.f$ for each vertex v
2: compute $G^\top$
3: call DFS($G^\top$), but in the main loop of DFS, consider the vertices in order of decreasing $u.f$ (as computed in line 1)
4: output the vertices of the tree in the depth-first forest formed in line 3 as a separate SCC

▶Strongly connected components: example

STRONGLY-CONNECTED-COMPONENTS(G)

1: call DFS(G) to compute finishing times $v.f$ for each vertex v
2: compute $G^\top$
3: call DFS($G^\top$), but in the main loop of DFS, consider the vertices in order of decreasing $u.f$ (as computed in line 1)
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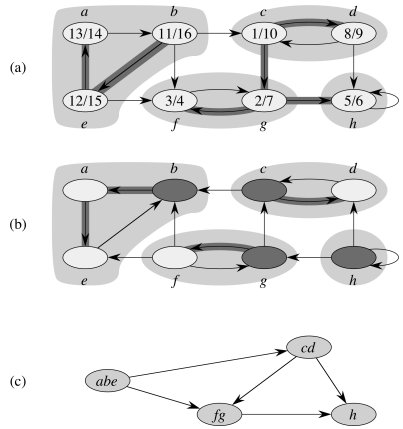
Runtime?

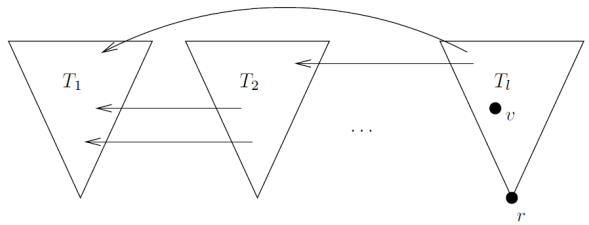


▶Correctness of the SCC algorithm

- Why on earth does this work? It's a miracle!
- Proof in the book is 3 pages of lemmas and not very intuitive.
- Let's use a simpler and more intuitive proof by Ingo Wegener:
- *A simplified correctness proof for a well-known algorithm computing strongly connected components*, Information Processing Letters 83(1), pages 17–19 (on Blackboard)

▶Correctness (2)

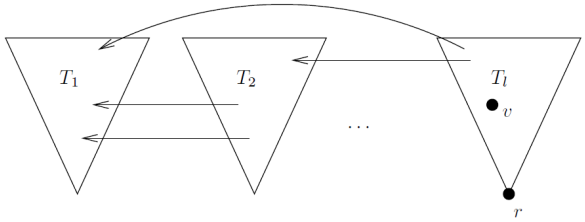
- Draw constructed depth-first trees from left to right and name them $T_1, T_2, \ldots, T_l$.
- Then edges between trees can only go right to left (otherwise, e.g. if there is an edge from T_1 to T_2, parts of T_2 would have been included in the depth-first tree T_1)



- Hence each SCC must be contained in one of the trees.

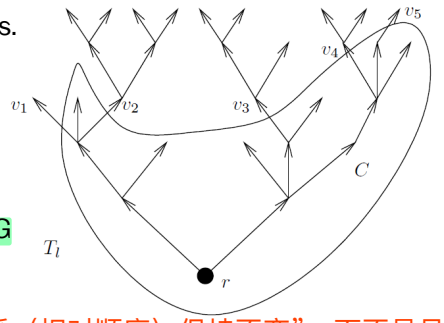
▶Correctness (3) – finding a first SCC

- The algorithm starts the second DFS on $G^\top$ computing the SCC C containing the **root r of the last tree** (as r finished last).
- We know that there is a path from r to all $v \in T_l$ (tree edges). So C is the set of all vertices v for which there is a path v to r in G. This is the set of **all vertices v reachable from r in $G^\top$**.
- After reversing all edges, DFS from r in $G^\top$ cannot leave T_l. Hence DFS in $G^\top$ from r outputs **exactly the SCC containing r**.



▶Correctness (4) – extracting a first SCC

- How does the SCC C containing r look like?
- If v belongs to C, then all vertices on the path r to v must also belong to C (as there is a path from v back to r).
- Hence C is a connected part of T_l.
- T_l without C splits into subtrees.
- $T_1, \ldots, T_{l-1}$ along with these subtrees is a depth-first forest which is also the result of a DFS traversal of G – C.
- The time stamps from DFS on G also work as time stamps for DFS on G – C! (**main insight**)

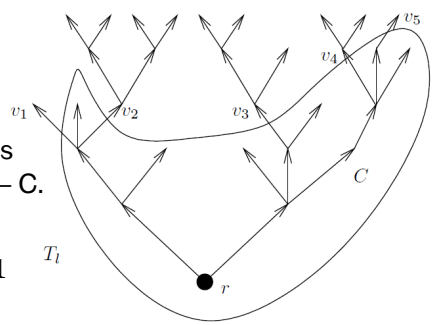


关键是“大小关系（相对顺序）保持不变”，而不是具体的数值。

▶Correctness (5) – repeated extraction

Proving correctness by induction over the number of SCCs:

- **Base case:** If the graph is a single SCC, the algorithm outputs it.
- Assume the algorithm is correct for graphs with $k-1$ SCCs.
- For a graph with k SCCs, the algorithm correctly outputs the SCC C containing the root r of the last DFS tree.
- Algorithm continues with vertices and depth-first (sub-)trees in G – C.
- By the induction hypothesis, it then outputs the remaining $k-1$ SCCs of G – C correctly as well.



▶Summary for Depth-First Search

- Depth-first search explores the graph going into depth and using backtracking in time $\Theta(|V| + |E|)$.
- DFS classifies edges into **tree, back, forward**, and **cross edges**.
- DFS is used to test whether a graph is **acyclic** in time $\Theta(|V| + |E|)$. Can be improved to $O(|V|)$ for undirected graphs (exercise!).
- DFS is used for **topological sorting** in directed acyclic graphs in time $\Theta(|V| + |E|)$.
- DFS is used to determine **strongly connected components** in graphs in time $\Theta(|V| + |E|)$.
- Seen detailed **correctness proofs** to demystify algorithms that appear magical at first glance.